Math 161
Example proof

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Here is an example of a complete proof, or rather two complete proofs, written in the way I requested for the first problem set. Either is acceptable: Proof 1 is more representative of the way we'll write proofs going forward, whereas Proof 2 is more formal.

Proposition (Distributivity of intersection over union)**.** *For any set X and any subsets $A, B, C \subset X$, we have*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Proof 1. To establish that these two sets are equal, we must prove each is contained in the other.

Suppose that $x \in A \cap (B \cup C)$. By the definition of intersection, this means that $x \in A$ and $x \in B \cup C$. By the definition of union, the latter means that $x \in B$ or $x \in C$. Therefore either $x \in A$ and $x \in B$, or $x \in A$ and $x \in C$. Applying the definitions of intersection and union again, we obtain $x \in (A \cap B) \cup (A \cap C)$. So we have shown that

$$A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C).$$

Now suppose that $x \in (A \cap B) \cup (A \cap C)$. By the definition of union, this means that $x \in A \cap B$ or $x \in A \cap C$. By the definition of intersection, the former means that $x \in A$ and $x \in B$, while the latter means that $x \in A$ and $x \in C$. In either case $x \in A$, and either $x \in B$ or $x \in C$. Thus (once more by the definitions of union and intersection) we have $x \in A \cap (B \cup C)$, and we have proved that

$$(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C),$$

as desired.

□

Proof 2. By the definitions of union and intersection, the proposition $x \in A \cap (B \cup C)$ means that

$$(x \in A) \text{ and } (x \in B \text{ or } x \in C),$$

while $x \in (A \cap B) \cup (A \cap C)$ means that

$$(x \in A \text{ and } x \in B) \text{ or } (x \in B \text{ and } x \in C).$$

In fact, we claim that for any propositions P, Q, R we have the logical equivalence

$$(P \text{ and } (Q \text{ or } R)) \iff ((P \text{ and } Q) \text{ or } (P \text{ and R})).$$

Taking P, Q, R to be the propositions $x \in A$, $x \in B$, and $x \in C$ respectively, this equivalence becomes the statement we are trying to prove.

To see this logical equivalence, we can compare truth tables.

P	T	T	T	T	F	F	F	F
Q	T	T	F	F	T	T	F	F
R	T	F	T	F	T	F	T	F
Q or R	T	T	T	F	T	T	T	F
P and (Q or R)	T	T	T	F	F	F	F	F

P	T	T	T	T	F	F	F	F
Q	T	T	F	F	T	T	F	F
R	T	F	T	F	T	F	T	F
P and Q	T	T	F	F	F	F	F	F
P and R	T	F	T	F	F	F	F	F
(P and Q) or (P and R)	T	T	T	F	F	F	F	F

This shows that for any given truth values of P, Q, R, the truth value of (P and (Q or R)) is the same as that of ((P and Q) or (P and R)), so these statements are indeed logically equivalent.

$\square$